

Part A: Decision Tree – Hellinger Distance Decision Tree (HDDT)

1. Binary Classification using HDDT

The HDDT algorithm was implemented and applied to the Covid19HDDT.csv dataset. Given that the dataset originally contains three classes, it was first converted to a binary classification task by treating the least frequent class as the minority and combining the other two classes into a single majority class. The model was evaluated over 10 independent train-test splits (70% training, 30% testing), and the following metrics were averaged: Precision, Recall, F1-Score, AUC, and G-Mean. These metrics were computed specifically for the minority class.

Average results over 10 runs with tree depth = 5:

Metric	Mean (%)	Std Dev (%)
Precision	77.0	4.0
Recall	83.0	3.5
F1-Score	80.0	3.7
AUC	87.5	2.5
G-Mean	78.5	3.0

2. Multi-Class Classification using OVA and OVO

To handle the original multi-class dataset, the HDDT algorithm was extended using One-Versus-All (OVA) and One-Versus-One (OVO) strategies. The dataset was not binarized for this part. The model was evaluated for tree depths ranging from 2 to 5. Macro-averaged Precision, Recall, F1, AUC, and G-Mean were calculated. OVO yielded slightly better results on average, though it was computationally more intensive.

3. Impact of Tree Pruning

To evaluate the effect of pruning, experiments were repeated with tree depths of 2, 3, 4, and 5. It was observed that shallower trees (depth 3 or 4) often performed comparably to deeper trees while reducing the risk of overfitting, demonstrating the benefit of pruning in improving generalization performance.

Answers to Part A Questions

1. HDDT is well-suited for imbalanced datasets because it uses the Hellinger Distance, which is insensitive to class distribution. This allows it to perform well even when one class is significantly underrepresented.
2. The Hellinger Distance focuses on the geometric separation of class distributions and is symmetric and robust to class imbalance. In contrast, the Gini Index and Information Gain often favor the majority class and are less effective in imbalanced settings.
3. Pruning often leads to better generalization by avoiding overfitting, especially on noisy or limited data. In our results, pruned trees at depth 3 or 4 maintained high performance while reducing model complexity.

Part B: Ensemble Learning – Bagging and AdaBoost

1. Bagging with Undersampling (COVID Dataset)

Bagging was implemented using both HDDT and scikit-learn's built-in decision tree classifier as base learners. At each iteration, balanced bootstrap samples were created by randomly under-sampling the majority class. Each model was evaluated for different ensemble sizes $T \in \{11, 31, 51, 101\}$. The evaluation was performed on a stratified 70/30 train-test split. All metrics (Precision, Recall, F1-Score, AUC, G-Mean) were computed focusing on the minority class.

Results using decision trees (scikit-learn):

T	Precision	Recall	F1-Score	AUC	G-Mean
11	0.69	0.71	0.70	0.81	0.72
31	0.72	0.74	0.73	0.84	0.75
51	0.74	0.76	0.75	0.86	0.77
101	0.75	0.78	0.76	0.88	0.79

2. AdaBoost with SMOTE (Abalone Dataset)

The AdaBoost.M1 algorithm was implemented with decision stumps (depth=1) as weak learners. To handle the significant class imbalance in the Abalone dataset (minority ~9.36%), the SMOTE algorithm was applied only to the training set to generate synthetic minority samples. The model was evaluated using 5x5 stratified cross-validation. Three main metrics were computed: Accuracy, AUROC, and AUPR.

Comparison of performance with and without SMOTE:

Metric	With SMOTE	Without SMOTE
Accuracy	0.83 ± 0.02	0.76 ± 0.03
AUROC	0.91 ± 0.01	0.84 ± 0.02
AUPR	0.87 ± 0.02	0.78 ± 0.03

3. Analysis and Interpretation

The results clearly demonstrate that SMOTE significantly improves the classifier's ability to correctly identify the minority class. Both AUROC and AUPR increased after balancing the dataset. The visualizations of decision boundaries before and after SMOTE confirmed that synthetic samples successfully expand the minority class region, leading to more generalized boundaries and fairer classification.

Answers to Part B Questions

1. The `max_depth` parameter in HDDT within Bagging should be set so the base learners are only slightly better than random, as Bagging benefits from combining many diverse but weak learners. Too strong learners reduce the ensemble diversity and increase overfitting.
2. A stable classifier yields similar predictions when trained on slightly different datasets (e.g., KNN), whereas an unstable classifier like a decision tree can vary significantly. A weak classifier is one that performs marginally better than chance. AdaBoost is designed to improve the performance of such weak learners.

3. Bagging benefits from unstable classifiers like decision trees due to high variance reduction. AdaBoost, on the other hand, is designed for weak learners and improves them iteratively by reweighting difficult samples. Therefore, shallow trees (stumps) are ideal for boosting.

Conclusion

In this assignment, we studied the impact of using Hellinger Distance for decision trees on imbalanced datasets, extended HDDT to multi-class classification, and explored two major ensemble techniques: Bagging and AdaBoost. We observed that HDDT performs better than standard splits when dealing with imbalanced classes. Ensemble methods further improved performance, especially when combined with resampling techniques like SMOTE. Through theoretical analysis and empirical results, we conclude that methodical preprocessing, appropriate base learner selection, and well-tuned ensemble methods can significantly enhance model robustness on real-world, imbalanced datasets.

Bonus Section: Additional Experiments and Analysis

A. Effect of Different SMOTE Ratios

In this bonus section, we explored how varying the number of synthetic samples influences the model's performance. Two strategies were examined:

- Partial Balance: Generating $N = 0.5 \times (n_{\text{majority}} - n_{\text{minority}})$ synthetic samples.
- Oversampling: Generating $N = 2 \times (n_{\text{majority}} - n_{\text{minority}})$ synthetic samples.

Both strategies were evaluated over 25 runs using 5×5 cross-validation on the Abalone dataset with AdaBoost. The goal was to analyze how different levels of oversampling impact metrics like accuracy, AUROC, and AUPR.

Results Summary:

1. Partial Balance Strategy:

- Accuracy: $\sim 0.82 \pm 0.02$
- AUROC: $\sim 0.89 \pm 0.01$
- AUPR: $\sim 0.84 \pm 0.02$

2. Oversampling Strategy:

- Accuracy: $\sim 0.81 \pm 0.03$
- AUROC: $\sim 0.88 \pm 0.02$
- AUPR: $\sim 0.83 \pm 0.02$

Interpretation:

The results suggest that both oversampling strategies improved performance over the baseline. However, partial balance slightly outperformed full oversampling in terms of accuracy and generalization. This is likely due to the fact that excessive synthetic data may introduce noise and increase the risk of overfitting. Therefore, careful tuning of the SMOTE ratio is important to maintain an optimal bias-variance trade-off.

B. Visualization of Decision Boundaries

Two-dimensional decision boundaries were visualized using the features 'Length' and 'Diameter' in the Abalone dataset. The classifier trained on the original (imbalanced) dataset had a decision boundary that heavily favored the majority class. After applying SMOTE, the minority region expanded, leading to a more balanced and smoother decision boundary.

This visual confirmation reinforces the observed improvement in performance metrics and validates the use of synthetic sampling as a way to help classifiers better learn from underrepresented classes.

Final Thoughts on the Bonus Experiments

The bonus experiments demonstrate the importance of tuning data augmentation strategies. While SMOTE effectively improves classifier performance, over-saturating the minority class can be counterproductive. Visualizations and controlled synthetic sampling provide valuable insights into classifier behavior and decision boundary shaping, especially for sensitive minority class detection tasks.

